

Primitive idempotents that cannot be homogenized in Koszul algebras

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September 9, 2026

Abstract

We construct primitive idempotents that no algebra automorphism can carry to a homogeneous idempotent in a locally finite standard graded algebra with semisimple degree zero. The algebra is a matrix algebra over a two-dimensional domain defined by two quadrics, and is Koszul. This gives a negative answer to Dramburg's Question 6.2, including over algebraically closed fields of characteristic zero. The same fixed algebra contains such an idempotent congruent to a constant matrix modulo any prescribed power of the positive-degree ideal. Each is conjugate to that constant matrix after completion. The proof combines an explicit matrix with a line-module obstruction, detected by its two trivializations over a Veronese ring and a polynomial quotient. Counterexamples exist over every field, in matrix size two outside characteristic two and size three in characteristic two.

1 The question and the construction

A nonnegatively graded k -algebra $A = \bigoplus_{j \geq 0} A_j$ is standard graded if it is generated by A_0 and A_1 , locally finite if each A_j is finite dimensional, and semiconnected if A_0 is semisimple. Dramburg asks whether every complete set of primitive orthogonal idempotents of such an algebra can be carried to its degree-zero parts by an automorphism [3, Question 6.2]. His earlier question explicitly includes the possibility of assuming that k is algebraically closed of characteristic zero [4]. We give counterexamples satisfying all these conditions, with the additional Koszul property.

The relation between idempotent matrices and projective modules is classical. Petersson describes line-module obstructions to automorphisms of two-by-two matrix algebras [6, Sections 5–6]. Kanwar, Leroy and Matczuk study conjugacy in polynomial and power-series extensions, and give a quaternion-polynomial example using Ojanguren–Sridharan [5, p. 9 of the author preprint]. De Jong also describes nonhomogeneous idempotents arising from line bundles in a Laurent extension [2]. Our contribution is an explicit split construction over every ground field, with a quadratic complete-intersection center, and an arbitrarily deep finite-jet failure in one fixed Koszul algebra. The underlying projective-module method is not new.

Fix a field k and put

$$R = k[a, b, c, z]/(b^2 - ac, c^2 - az), \quad \deg a = \deg b = \deg c = \deg z = 1. \quad (1)$$

For $r \geq 0$, define four elements of R by

$$u_{2,r} = cz^{2r+1}, \quad u_{3,r} = bz^{3r+2}, \quad u_{4,r} = az^{4r+3}, \quad u_{5,r} = bc z^{5r+3}, \quad (2)$$

and set

$$p_r = \begin{pmatrix} 1 - u_{4,r} & u_{2,r} + u_{3,r} + u_{4,r} + u_{5,r} \\ u_{2,r} - u_{3,r} & u_{4,r} \end{pmatrix}. \quad (3)$$

Write E_{ij} for the standard matrix units. In $A = M_n(R)$ let

$$e_r = \text{diag}(p_r, 0_{n-2}), \quad f_r = \text{diag}(1 - p_r, 0_{n-2}), \quad A_j = M_n(R_j), \quad A_+ = \bigoplus_{j>0} A_j.$$

The notation for the zero block is omitted when $n = 2$.

Theorem 1. *Let $n \geq 2$ be an integer whose image in k is nonzero. For the ring and matrices above, the following statements hold.*

1. *R is a two-dimensional standard graded domain and a quadratic complete intersection. The algebra A is locally finite, standard graded and Koszul, with $A_0 = M_n(k)$.*
2. *For every $r \geq 0$, the elements $e_r, f_r, E_{33}, \dots, E_{nn}$ form a complete set of primitive orthogonal idempotents. Their degree-zero parts are $E_{11}, E_{22}, E_{33}, \dots, E_{nn}$.*
3. *No k -algebra automorphism of A carries e_r to any homogeneous idempotent. In particular, no automorphism carries the displayed complete system to its degree-zero parts, even allowing a permutation.*
4. *We have $e_r - E_{11} \in A_+^{2r+2}$. Nevertheless e_r and E_{11} are conjugate by a unit congruent to 1 modulo A_+^{2r+2} in the A_+ -adic completion. Their images are conjugate in every quotient A/A_+^N , $N \geq 1$.*

Taking $n = 2$ when $\text{char } k \neq 2$, and $n = 3$ otherwise, gives a counterexample over every field. In particular, $n = 2$ works over $\mathbb{C}$. The theorem concerns arbitrary algebra automorphisms, not just conjugation by units or automorphisms fixing the center pointwise.

2 The center and the auxiliary rings

Lemma 2. *The map in (1) induced by*

$$a \mapsto s^4, \quad b \mapsto s^3t, \quad c \mapsto s^2t^2, \quad z \mapsto t^4$$

identifies R with the indicated subring of $k[s, t]$. The two defining quadrics form a Gröbner basis. The ring R is Koszul.

Proof. Use lexicographic order $b > c > a > z$. The leading monomials b^2 and c^2 are relatively prime, so Buchberger's criterion gives the asserted Gröbner basis. Its standard monomials are

$$a^i z^j b^\epsilon c^\delta, \quad i, j \geq 0, \quad \epsilon, \delta \in \{0, 1\}. \quad (4)$$

Their images in $k[s, t]$ have exponent pairs

$$(4i + 3\epsilon + 2\delta, 4j + \epsilon + 2\delta).$$

The first exponent modulo 4 determines (ϵ, δ) , since the four possibilities are 0, 3, 2, 1. The two exponents then determine i, j . Thus the standard monomials have distinct images, proving injectivity and the domain assertion.

Formula (4) also makes R a free module of rank four over $k[a, z]$, with basis $1, b, c, bc$. Consequently $\dim R = 2$, and the defining ideal has height two. Two generators of height two in a polynomial ring form a regular sequence, so this is a complete intersection. Finally, a standard graded algebra with a quadratic Gröbner basis is Koszul; see, for example, [1, Section 1]. $\square$

Work henceforth in $k[s, t]$, using the identifications of Lemma 2. Introduce

$$B = k[s^4, s^3t, s^2t^2, st^3, t^4], \quad d = st^3, \quad C = k[z]. \quad (5)$$

Thus B is the fourth Veronese subring. Give its degree-four monomials degree one. Let J be the k -span in B of all its monomials having s -exponent at least two. This is an ideal. In the quotient only the monomials z^j and dz^j , $j \geq 0$, remain. Hence

$$D = B/J \cong k[z, d]/(d^2). \quad (6)$$

The symbol d in D denotes the residue class of st^3 .

The inclusion $R \rightarrow B$ followed by the quotient to D sends a, b, c to zero and z to z . It therefore agrees with the composition $R \rightarrow C \rightarrow D$, where $R \rightarrow C$ kills a, b, c . We will use this commutative square of ring maps. We do not need flatness of B over R , a computation of $\text{Pic}(B)$, or a description of the full normalization of R .

Both B and C have unit group $k^\times$. For B , this follows because it is a connected positively graded domain: the highest nonzero degrees of two nonzero elements add under multiplication, so a product equal to 1 forces both elements to have degree zero. The assertion for $C = k[z]$ is immediate.

3 An explicit line-module obstruction

For an indeterminate x , consider the matrix

$$T(x) = \begin{pmatrix} 1 + x + x^2 + x^3 & -x^2 \\ x^2 & 1 - x \end{pmatrix} \in M_2(\mathbb{Z}[x]). \quad (7)$$

Its determinant is $(1 - x)(1 + x + x^2 + x^3) + x^4 = 1$. Direct multiplication gives

$$T(x)E_{11}T(x)^{-1} = \begin{pmatrix} 1 - x^4 & x^2 + x^3 + x^4 + x^5 \\ x^2 - x^3 & x^4 \end{pmatrix}. \quad (8)$$

Put $\xi_r = dz^r \in B$. The identities

$$\xi_r^2 = cz^{2r+1}, \quad \xi_r^3 = bz^{3r+2}, \quad \xi_r^4 = az^{4r+3}, \quad \xi_r^5 = bc z^{5r+3}$$

show that (8), evaluated at ξ_r , is precisely p_r and has entries in R . Since $R \rightarrow B$ is injective, p_r is idempotent over R .

Set $L_r = p_r R^2$. The determinant and trace of p_r are 0 and 1. Over every residue field its image has dimension one; thus L_r and $(1 - p_r)R^2$ are finitely generated projective modules of rank one. Here a line module means a projective module of rank one.

Proposition 3. *For every $r \geq 0$ and every positive integer m nonzero in k , the line module $L_r^{\otimes m}$ is not free. The line modules L_r , $r \geq 0$, are pairwise nonisomorphic.*

Proof. After base change to B , the module L_r has the basis vector

$$v_r = T(\xi_r) \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Indeed, conjugation in (8) identifies the image with the span of the first column of an invertible matrix. After base change to C , the idempotent becomes E_{11} , so $L_r \otimes_R C$ has basis $w = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

These identifications are compatible over D . Since $\xi_r^2 = 0$ in D , we have

$$T(\xi_r) \bmod J = \text{diag}(1 + dz^r, 1 - dz^r), \quad v_r = (1 + dz^r)w \quad \text{over } D. \quad (9)$$

Tensor powers of these basis vectors consequently differ by $(1 + dz^r)^m$.

Suppose $L_r^{\otimes m} \cong R$ and choose a global basis. Over B its expression is $\beta v_r^{\otimes m}$ for some $\beta \in B^\times$; over C it is $\gamma w^{\otimes m}$ for some $\gamma \in C^\times$. These scalars must be units because both expressions are bases. They lie in $k^\times$, by the unit computations above. Compatibility over D gives

$$\beta(1 + dz^r)^m = \gamma.$$

The left side is $\beta(1 + mdz^r)$. The elements 1 and dz^r are linearly independent over k , and $m \neq 0$ in k , a contradiction.

Likewise, an isomorphism $L_r \cong L_s$ becomes multiplication by units in the two chosen trivializations. Compatibility implies an equality

$$\beta(1 + dz^r) = \gamma(1 + dz^s), \quad \beta, \gamma \in k^\times.$$

Comparing the constant terms gives $\beta = \gamma$, and then $z^r = z^s$ in $k[z]$, so $r = s$.

All base changes used here are applied to direct summands of free modules, and to a hypothesized isomorphism. They require no flatness assumption. $\square$

In characteristic zero, each $[L_r]$ has infinite order in $\text{Pic}(R)$. We have not computed this Picard group, and need no assertion about its other elements.

4 Excluding arbitrary automorphisms

The following matrix-unit argument is a useful form of the classical projective-module obstruction. It avoids an assumption that an automorphism fixes the center.

Lemma 4. *Let S be a commutative ring and let $e \in M_n(S)$ be an idempotent whose image $P = eS^n$ is a line module. If a ring automorphism of $M_n(S)$ carries e to E_{11} , then $P^{\otimes n} \cong S$.*

Proof. Let α be such an automorphism and set $q_{ij} = \alpha^{-1}(E_{ij})$. These matrices satisfy the matrix-unit identities, and $q_{11} = e$. They yield a direct sum

$$S^n = \bigoplus_{i=1}^n q_{ii} S^n.$$

Multiplication by q_{i1} is an S -linear isomorphism from P to $q_{ii} S^n$, with inverse multiplication by q_{1i} . Thus $S^n \cong P^{\oplus n}$. Taking the top exterior power gives $S \cong P^{\otimes n}$.

The matrices q_{ij} act S -linearly regardless of the action of α on scalar matrices. In particular, the proof applies to automorphisms acting nontrivially on the center. $\square$

Proof of Theorem 1. The ring statements follow from Lemma 2. The entrywise grading on A is standard and locally finite, and $A_0 = M_n(k)$ is semisimple. For the Koszul assertion, take a linear graded projective resolution of k over R and apply the exact graded Morita functor

$$M_n(R)E_{11} \otimes_R -.$$

It gives a linear resolution of the simple $M_n(k)$ -module k^n , regarded as an A -module. Taking n copies gives a linear resolution of A_0 as a left A -module, which is the Koszul condition with semisimple degree zero.

The identities for p_r give the stated orthogonal decomposition of 1. Every displayed summand has image of rank one over R . For a line module P one has $\text{End}_R(P) \cong R$, as can be checked after localization. Since R is a domain, this endomorphism ring has no nontrivial idempotents. Hence each displayed matrix idempotent is primitive. The degree-zero assertions follow directly from (2).

The image of e_r is isomorphic to L_r . Proposition 3 says that $L_r^{\otimes n}$ is not free. Lemma 4 therefore excludes an automorphism taking e_r to E_{11} . A nonzero homogeneous idempotent in a nonnegative grading has degree zero, since its square has twice its degree. A primitive idempotent in $M_n(k)$ has rank one: any larger rank splits into orthogonal constant idempotents in A . Every constant rank-one idempotent is conjugate over k to E_{11} . Composing with that constant conjugation proves the third assertion, including the statement about permutations of the complete system.

The four elements $u_{j,r}$ have respective degrees $j(r+1)$ for $2 \leq j \leq 5$. Thus $e_r - E_{11} \in A_{\geq 2r+2} = A_+^{2r+2}$, where equality uses the standard grading. Put $E = E_{11}$ and

$$W = e_r E + (1 - e_r)(1 - E) = 1 + (e_r - E)(2E - 1). \quad (10)$$

Then $WE = e_r W$, by idempotence. Also $W - 1 \in A_+^{2r+2}$. Therefore the geometric series for $(1 + (W - 1))^{-1}$ converges in the A_+ -adic completion, and becomes a finite inverse in every quotient A/A_+^N . In each such ring $e_r = WEW^{-1}$, proving the final assertion. $\square$

Corollary 5. *For the fixed algebra A in Theorem 1 and every $N \geq 1$, there is a primitive idempotent congruent to E_{11} modulo A_+^N whose orbit under all algebra automorphisms contains no homogeneous idempotent. The family e_r consists of pairwise distinct inner-conjugacy classes.*

Proof. Choose r with $2r + 2 \geq N$. For the last assertion, an inner conjugation between e_r and e_s would induce an R -linear isomorphism between their images L_r and L_s , contradicting Proposition 3 when $r \neq s$. $\square$

5 Scope and consequences

The construction gives a negative answer to the homogenization question even for a prime algebra finite over its center, with center a two-dimensional quadratic complete-intersection domain. The obstruction is algebraic: completion removes it, and no fixed finite order of agreement with the degree-zero part detects it. This does not assert that the e_r lie in distinct orbits under all automorphisms; we only need to exclude the homogeneous orbit for each of them.

There is no conflict with the uniqueness theorem for standard semiconnected locally finite gradings [3, Theorem 1.2]. That theorem compares gradings already given on an algebra. The primitive systems here cannot be made homogeneous by choosing another grading of that class: a graded isomorphism from such a grading to the present one would carry e_r to a homogeneous idempotent, contrary to Theorem 1.

The center in (1) is not seminormal: d belongs to its fraction field, since $d = bc/a$, and $d^2, d^3 \in R$ while $d \notin R$. This failure provides the nonconstant nilpotent unit $1 + dz^r$ in the comparison ring D . The example therefore does not settle analogous questions restricted to normal centers or to algebras of finite global dimension. These additional restrictions require separate arguments.

References

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